

Opioid Overdose with ARDS



Section I: Scenario Demographics

Scenario Title:	Opioid Overdose with Acute Respiratory Distress Syndrome (ARDS)
Date of Development:	23/03/2015
Target Learning Group:	☐ Juniors (PGY 1 - 2) ☐ Seniors (PGY $\geq$ 3) ☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Martin Kuuskne
Affiliations/Institution(s):	McGill University
Contact E-mail (optional):	martin.kuuskne@mail.mcgill.ca

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To recognize and treat opioid toxicity and one of its rare respiratory complications.
CRM Objectives:	1) Distribute the workload effectively during the intubation of a critically ill patient. 2) Reevaluate the case and avoid fixation error by verbalizing a differential diagnosis.
Medical Objectives:	1) Recognize the clinical manifestations of a severe opioid toxicity through a thorough physical examination 2) Employ a wide differential diagnosis in the treatment of an intoxicated patient with respiratory failure.

Case Summary: Brief Summary of Case Progression and Major Events
A 34-year-old male was found unconscious in an alleyway by bystanders who called EMS. The patient presents with a clinical opioid intoxication requiring naloxone administration. The patient also presents with acute respiratory distress syndrome (ARDS) secondary to heroin use requiring airway support, intubation and mechanical ventilation.

References
Nelson, L., Lewin, N., Howland, M., et al. Goldfrank's Toxicologic Emergencies. 9 th edition. 2011 McGraw-Hill Companies.
Sporer, K., & Dorn, E. (2001). Heroin-Related Noncardiogenic Pulmonary Edema : A Case Series. <i>Chest</i> , 120(5), 1628-1632.



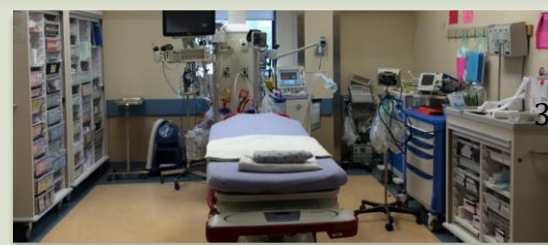
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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
You are working in a community centre emergency department. A 34-year-old male is being brought into the resuscitation bay by EMS after being found unconscious in an alley-way by bystanders who called 911. The patient was given O ₂ by facemask and no other therapies en-route.			
B. Scenario Cast & Realism			
Patient:	☐ Computerized Mannequin	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual
	☒ Mannequin		☐ Physical
	☐ Standardized Patient		☒ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A
Confederates	Brief Description of Role		
Nurse	Confirms exam findings as listed in physical exam section.		
C. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☒ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
E. Moulage			
Syringes, a spoon and a lighter in a plastic bag which can be hidden in the patient's pocket			
F. Approximate Timing			
Set-Up:	10 min	Scenario:	12 min
		Debriefing:	15 min

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Section V: Patient Data and Baseline State

A. Patient Profile and History			
Patient Name: Collin Osborne		Age: 34	Weight: 100kg
Gender: ☒ M ☐ F		Code Status: Unknown	
Chief Complaint: Altered Level of Consciousness			
History of Presenting Illness: The patient was found unconscious in an alleyway by bystanders. There was vomit around the patient when found. No friends or family could be reached. The patient was "stable" en-route to the hospital. The patient was given O ₂ by facemask and no other therapies en-route.			
Past Medical History:	Unknown	Medications:	Unknown
Allergies: Unknown			
Social History: Unknown			
Family History: Unknown			
Review of Systems: Unable to obtain ROS from patient.			
B. Baseline Simulator State and Physical Exam			
☐ No Monitor Display		☒ Monitor On, no data displayed	☐ Monitor on Standard Display
HR: 55/min	BP: 100/65	RR: 4/min	O ₂ SAT: 79%
Rhythm: Sinus	T: 34.5°C	Glucose: 6.7mmol/L	GCS: 3 (E1 V1 M1)
General Status: Non-responsive			
CNS:	Pupils 1mm bilaterally, non-reactive, limp extremities, decreased DTRs diffusely, downgoing plantar reflexes.		
HEENT:	Normal		
CVS:	Normal S1 + S2, no extra heart sounds		
RESP:	Diffuse crackles bilaterally		
ABDO:	Soft, non-tender		
GU:	Decreased bowel sounds		
MSK:	Normal	SKIN:	Track marks in right AC fossa.

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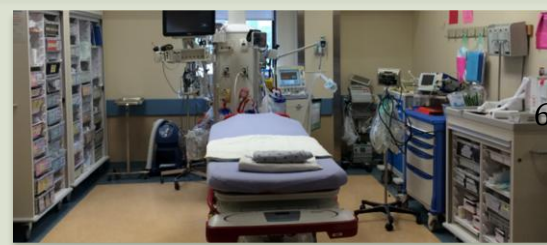
Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus HR: 55/min BP: 100 / 65 RR: 4/min O ₂ SAT: 79% T: 34.5°C	Decreased mental status, GCS 3 Pulse present. Pupils 1mm bilaterally.	<u>Learner Actions</u> ☐ Attaches monitors ☐ Acquires IV access ☐ Supplemental O ₂ ☐ Hx/Physical Exam ☐ EKG/CXR ☐ Lab investigations ☐ Recognizes opioid toxidrome ☐ Naloxone IV bolus #1	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - O ₂ by facemask → O ₂ SAT to 84% - O ₂ by BVM → O ₂ SAT to 88% <u>Triggers</u> <i>For progression to next state</i> - Naloxone bolus # 1 → 2. Acute Pulmonary Edema - 5 minutes → 2. Acute Pulmonary Edema
2. Acute Pulmonary Edema → HR: 60/min → BP: 109/71 → RR: 8/min → O ₂ SAT: 88% → Trending to 70% over 2 minutes	Increased mental status GCS: 7 E1none V2sounds M4 withdraws Pupils 2 mm bilaterally	<u>Learner Actions</u> ☐ Warm IV NS bolus or drip ☐ Naloxone IV bolus #2 ☐ Intubation ☐ Optimizes preoxygenation ☐ Apneic oxygenation ☐ Induction med ☐ Intubation paralytic med	<u>Modifiers</u> - Naloxone IV bolus #2 or Drip → RR to 14, pupils to 4mm bilat and reactive - Preoxygenation with BiPAP or assisted ventilation with BVM+PEEP Valve → O ₂ SAT to 90% <u>Triggers</u> - Intubation → 3. Intubation
3. Intubation → HR: 90 → BP: 110/70 → O ₂ SAT: 92% → RR: set Display FiO ₂ = 1.0 T _v = 500 PEEP = 5 T _v = 500 mL ETCO ₂ = 60	Pt paralyzed post intubation	<u>Learner Actions</u> ☐ Confirmation of tube ☐ NG tube ☐ Foley Catheter ☐ Post intubation CXR ☐ Naloxone IV Drip ☐ Antibiotics (Ceftriaxone & Vancomycin) ☐ Bair Hugger ☐ Lung Protective Settings (T _v =8mL/kg PBW, P _{pl} ≤30)	<u>Modifiers</u> - Bair Hugger → T to 36.0°C <u>Triggers</u> - 3 minutes into state → 4. Resolution
4. Resolution	Pt paralyzed and sedated post intubation	<u>Learner Actions</u> ☐ ICU consultation ☐ Poison Centre Consultation ☐ Re-evaluation of case ☐ CT Head	END SCENARIO PRN

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results					
VBG	pH: 7.2	PCO ₂ : 75	PO ₂ : 45	HCO ₃ : 24	Lactate: 1.9
Images (ECGs, CXRs, etc.)					
Pre intubation CXR http://www.radiology.vcu.edu/programs/residents/quiz/pulm_cotw/PulmonConf/09-03-04/68yM%2008-03-04%20CXR.jpg			Post intubation CXR http://courses.washington.edu/med620/images/my_c3fig1.jpg		
12-Lead EKG EKG: http://www.emedu.org/ecg/images/sb_1a.jpg					
Ultrasound Video Files (if applicable)					
Lung Ultrasound			Cardiac Ultrasound: Grossly normal EF		

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Sample Questions for Debriefing	
1) What are the proposed mechanisms of opioid induced ARDS? 2) What are key leadership priorities during an intubation? Can you briefly hand over leadership to the intubator? 3) What are ways of optimally pre-oxygenating a patient prior to intubation? 4) What other differential diagnoses are important to consider in this case? 5) Do you have an approach to naloxone dosing? If so, what do you use? 6) What were some of the teamwork challenges faced during this case? 7) What were some of the communication challenges faced during this case?	
Key Moments	
Recognition of opioid toxidrome	
Recognition of ARDS	
Intubation	

